The Agony and the Ecstasy

Constructing a “Crash-Filtered” Equity Index using Machine Learning

Tristan Leiter

Vienna University of Business and Economics

Supervisor: Prof. Kurt Hornik

May 25, 2026

The Agony and Ecstasy

Passive indices are exposed to both extremes

Long-run index returns are driven by a few high-performing stocks, while a considerable share of constituents suffer catastrophic declines.

Ecstasy

- Extreme winners carry a large share of long-run wealth creation.
- Winners can be volatile, expensive, or distressed.
- Thus, even a small amount of disregarded winners can hurt performance.

Agony

- Some constituents experience deep and persistent drawdowns without recovery.
- Passive strategies may hold imploding firms for too long.

Solution

The challenge is not to eliminate highly volatile stocks. It is to distinguish recoverable volatility from persistent implosion. The empirical question is whether catastrophically declining firms are forecastable using ideas from credit-risk modelling, but with market-based distress signals.

Research Question

Main research question

To what extent does a "Crash-Filtered" equity index, constructed via probabilistic implosion modeling, generate superior risk-adjusted returns compared to the market benchmark and traditional minimum-volatility strategies?

The probabilistic implosion model follows the Catastrophic Stock Implosion approach proposed by Tewari et al. (2024). Instead of relying only on formal bankruptcy declarations, CSI uses market-based price-path signals.

Subquestions:

- How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?
- To what extent do Ensemble methods reduce the False Positive Rate (classifying recoverable volatility as CSI) compared to traditional volatility-based exclusion strategies, while maintaining Recall?
- Which features are most important for distinguishing between "Zombie" firms (CSI) and non-zombie firms?

Dataset

Empirical design

The final panel keeps only observable CRSP firm-years: each row has the required monthly return and market-cap scaffold. Temporary and permanent CSI use the same 188,460-row scaffold; unresolved $y = \mathrm{NA}$ rows are retained in the panel but excluded from supervised model metrics.

Data sources

CRSP
monthly returns, market cap, delistings, wealth paths

Compustat
annual accounting fundamentals

FRED
macro variables

Universe

Minimum size
market cap $\geq$ USD 100M

Observable scaffold
188,460 annual firm-year rows in both tracks

Index buckets
Total, Large, Mid, Small Cap

Sample period

Coverage
Jan 1993 – Dec 2024 (32y, monthly frequency)

Supervised labels
train/test labelled; OOS may retain unresolved labels

metrics use labelled rows only

Temporal split

Train/CV	Test	OOS
143,173 rows	18,111 rows	27,176 rows

1993 — 2015 — 2019 — 2024

Methodology I: Response Variable Classification

Following Tewari et al. (2024), a stock is classified as a CSI candidate if it satisfies all three path criteria:

- **Initial Crash (C):** more than a C drawdown from the trailing peak marks the beginning of the "zombie" period. $C \leq -80\%$
- **Non-Recovery (M):** maximum cumulative return remains below the recovery ceiling M during the zombie period. $M \leq -20\%$
- **Zombie Period (T):** duration of the confirmation window during which non-recovery must hold. $T = 18$ months

The prediction task is annual. A trigger in calendar year $t + 1$ gives the firm-year observation at t a positive label when the track-specific confirmation rule is met:

$$y_{i,t} = \begin{cases} 1, & \text{if stock } i \text{ triggers a qualifying CSI event within } [t, t+h], \\ 0, & \text{otherwise.} \end{cases}$$

Two final response tracks

Temporary CSI enables the accepted terminal-failure route and keeps censored triggers plus unavailable 2025 event-year labels as $y = \mathrm{NA}$. **Permanent CSI** uses event-window-only unresolved PCL windows; broad no-trigger calendar censoring is not used.

Applying the classification: Temporary CSI

Temporary CSI

Terminal-failure route enabled. The temporary-CSI label has 8,517 positive annual cells in the full observable scaffold. The 8,674 OOS $y = \mathrm{NA}$ rows are unresolved labels: 3,342 censored trigger labels plus 5,332 observable 2024 rows that would require unavailable 2025 event-year labels.

Split	Obs. rows	$y = 0$	$y = 1$	$y = NA$	Labelled	Prev.
Full	188,460	171,269	**8,517**	8,674	179,786	4.74%
Train	143,173	136,630	6,543	0	143,173	4.57%
Test	18,111	17,307	804	0	18,111	4.44%
OOS	27,176	17,332	1,170	8,674	18,502	6.32%

Motivation for robustness

Prevalence is computed on labelled cells only: $y = 1/(y = 0 + y = 1)$. The retained $y = \mathrm{NA}$ rows are observable firm-years with unresolved labels, not dropped rows; they are excluded from supervised training and label-based evaluation.

Applying the classification: Permanent CSI

Permanent CSI

Permanent CSI uses the same 188,460 observable firm-year scaffold. Unresolved labels are limited to event-window PCL cases; broad no-trigger calendar censoring is explicitly not used.

Split	Obs. rows	$y = 0$	$y = 1$	$y = NA$	Labelled	Prev.
Full	188,460	181,368	**6,258**	834	187,626	3.34%
Train	143,173	137,794	5,379	0	143,173	3.76%
Test	18,111	17,511	542	58	18,053	3.00%
OOS	27,176	26,063	337	776	26,400	1.28%

Metric denominator

For both tracks, labelled rows are $y = 0 + y = 1$ and prevalence is $y = 1/(y = 0 + y = 1)$. Unresolved rows remain in the cleaned panel but are excluded from supervised training and evaluation metrics.

Robustness I: Temporary CSI Sensitivity Grid

Completed local sensitivity evidence

The temporary-CSI C/M/T sensitivity grid is available locally. All 27 run IDs are represented; 24 have complete/reused compact results and 3 are explicitly marked `blocked_partial`. Derived presentation summaries remain local-only under `03_Data_Output`.

Grid design

- Crash threshold C: -60%, -80%, -90%.
- Recovery threshold M: 0%, -20%, -30%.
- Confirmation window T: 12, 18, 28 months.
- Track: temporary CSI only; raw model sensitivity.

Local status

Status	Runs
Represented in registry	27
Complete or safely reused	24
`blocked_partial`	3
Checksum failures	0

Blocked partial runs

`C080_M000_T012`, `C080_M000_T018`, and `C060_M020_T028` are listed as `blocked_partial`; they are not treated as completed evidence.

Remark: Sources: local `presentation_ready/sensitivity_cmt_model_summary.csv`; `temporary_blocked_config_disclosure.csv`; AE-SENS-PRES-003 evidence.

Methodology II: “Temporary-CSI”

Revision

Original CSI rule unchanged. Temporary CSI uses the accepted `CSI_POSITIVE_EVENT_STATUSES`; a terminal-failure route is added only after a valid drawdown trigger. Censored triggers and unavailable 2025 event-year labels are retained as $y = \mathrm{NA}$.

Drawdown trigger $D_{i,m} \le C$ → Confirmation window T months → $y = 1$ CSI, no recovery to M

Drawdown trigger $D_{i,m} \le C$ → Confirmation window $\mathit{trigger} < \mathit{dlstdt} \le \mathit{trigger} + T$ → $y = 1$ TERMINAL FAILURE, $\mathit{dlstcd} \in \{572, 573, 574\}$

Scenario	Bankrupting	Detected	Missed	**Detection**
Before: `CSI` only	629	151	478	24.0%
After: `CSI` + terminal failure	629	**545**	84	**86.65%**

The 84 remaining misses are treated as methodological exclusions under the retained CSI-trigger rule, not as known pipeline errors.

Robustness II: Sensitivity Results and Limits

Sensitivity winners point to different objectives, not one universal setting

Objective	Config	CV AP	CV AUC	CV FPR3	Test AP	Test AUC	Test FPR3
Composite	C090_M000_T012	0.442	0.916	0.370	0.438	0.917	0.347
AP winner	C060_M000_T012	**0.473**	0.825	0.171	**0.476**	0.859	0.173
11C TM	C090_M020_T018	0.198	0.897	0.279	0.209	0.906	0.271
Baseline	C080_M020_T018	0.203	0.862	0.230	0.199	0.877	0.200

Interpretation

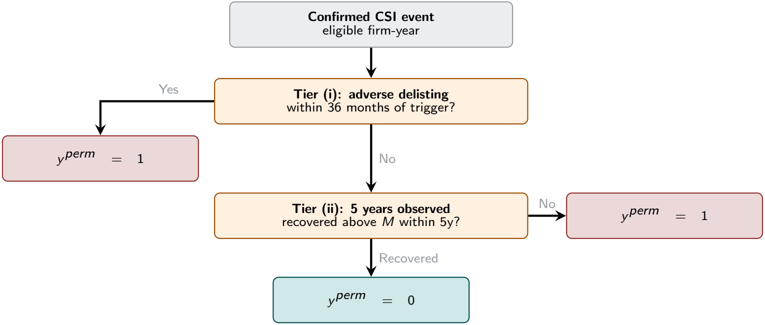
The baseline remains defensible as a continuity setting, but it is not the top-ranked sensitivity configuration. CV/test metrics show that the AP winner is not the index-alpha winner, so predictive ranking and portfolio performance are related but distinct objectives.

Use sensitivity as robustness evidence, not as a replacement for the final index suite. OOS index-alpha comparisons are shown later in the index and robustness sections. *Remark:* Sources: local `sensitivity_cmt_model_summary.csv`; `temporary_presentation_key_findings.csv`. FPR3 columns are R@FPR3.

Methodology III: “Permanent-CSI”

Dealing with recovering firms

Permanent-CSI starts from a confirmed CSI event and separates terminal failure, durable non-recovery, resolved recovery, and unresolved PCL windows. Its $y = \mathrm{NA}$ policy is event-window-only; broad no-trigger calendar censoring is not used.



Contrast with temporary CSI

Temporary CSI applies a finite lockout after a flag. Permanent-CSI marks terminal failure or durable non-recovery as $y^{perm} = 1$, resolved recovery as $y^{perm} = 0$, and unresolved PCL windows as $y = \mathrm{NA}$. The final full panel has 6,258 positives across 187,626 labelled firm-years (3.34%).

Dataset: Descriptive Statistics

CSI positives are rare and economically distinct from non-positives

Both label definitions isolate a small distressed tail. The key contrast is not only prevalence: positive firms are much smaller and show worse realized returns and profitability.

Sample	CSI	y	Obs.	Share	Firms	Median mcap	Ann. ret.	ROA
Full	Temp.	0	171,269	95.26%	19,073	254.9	4.9%	1.6%
Full	Temp.	1	**8,517**	4.74%	5,603	**74.9**	**-35.0%**	**-18.8%**
Full	Perm.	0	181,368	96.66%	19,565	252.6	4.0%	1.5%
Full	Perm.	1	6,258	3.34%	4,696	**65.7**	**-37.5%**	**-20.0%**

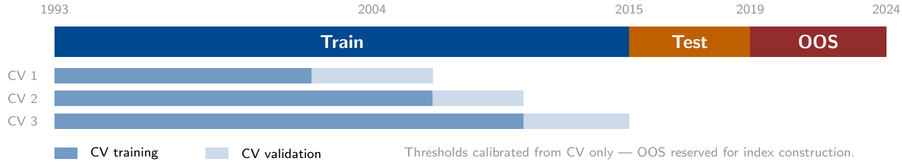
Read

The full temporary-CSI classification contains **8,517** positive annual cells. Positive labels identify much smaller firms with sharply negative returns and profitability.

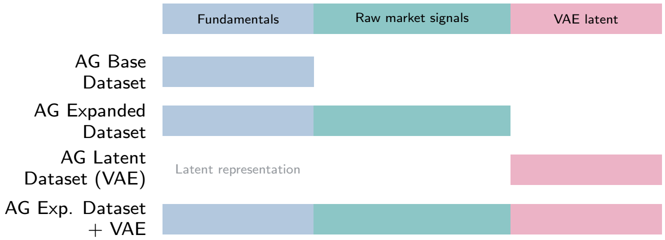
Remark: Obs. are annual label cells; Firms are unique firms. Market cap is USD millions.

Modelling I: Setup and Feature Engineering

Validation design



Final feature-set keys



Modelling II: Results Temporary CSI

Main read

Temporary CSI remains highly rankable, but AP and fixed-FPR recall are the decision metrics because positives are rare. **AG Exp. Dataset + VAE** is strongest on train/CV and OOS AP/FPR recall; **AG Expanded Dataset** remains the strongest 2016–2019 test benchmark and has marginally higher OOS AUC.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.2046	0.8635	0.2344	**0.1985**	**0.8766**	**0.2002**
AG Base Dataset	0.1923	0.8446	0.2206	0.1656	0.8538	0.1517
AG Latent Dataset (VAE)	0.1659	0.8454	0.1706	0.1501	0.8438	0.1294
AG Exp. Dataset + VAE	**0.2114**	**0.8667**	**0.2478**	0.1864	0.8741	0.1779

Interpretation

VAE latent information helps most when added to the expanded feature set and evaluated OOS. The expanded-dataset benchmark remains competitive, so model slides alone do not establish index-level superiority; the index section evaluates economic performance.

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A10.

Modelling III: Results Permanent-CSI

Main read

Permanent CSI is harder: positives are rarer and OOS prevalence is much lower. **AG Exp. Dataset + VAE** is the best non-raw CV/test challenger, while **AG Latent Dataset (VAE)** is the strongest OOS robustness signal.

Model	CV-AP	CV-AUC	CV-FPR3	Test-AP	Test-AUC	Test-FPR3
AG Expanded Dataset	0.1854	**0.8757**	**0.2607**	**0.1416**	0.8810	0.1808
AG Base Dataset	0.1785	0.8677	0.2379	0.1289	0.8627	0.1697
AG Latent Dataset (VAE)	0.1426	0.8460	0.1960	0.1115	0.8496	0.1513
AG Exp. Dataset + VAE	**0.1883**	0.8719	0.2578	0.1415	**0.8838**	**0.2011**

Interpretation

AG Expanded Dataset remains a strong reporting baseline. For permanent CSI, AG Exp. Dataset + VAE is the primary non-raw challenger, while AG Latent Dataset (VAE) is retained for OOS robustness rather than treated as a clean replacement.

Remark: CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A11.

Modelling IV: What VAE Features Add

Feature-set interpretation

The VAE does not replace expanded-dataset predictive information. It is useful as an augmentation or robustness signal: AG Exp. Dataset + VAE improves temporary OOS AP/FPR recall, while AG Latent Dataset (VAE) is the strongest permanent OOS model.

Temporary CSI

- AG Exp. Dataset + VAE leads OOS AP: 0.3152.
- OOS R@FPR1/3/5 improves to 0.1051 / 0.2632 / 0.4128.
- AG Expanded Dataset still has the slightly higher OOS AUC: 0.8961 vs 0.8950.

Permanent CSI

- AG Exp. Dataset + VAE leads test AUC and FPR3/5 recall.
- AG Latent Dataset (VAE) leads OOS AP/AUC/Brier/FPR recall.
- AG Base Dataset remains a useful fundamentals-only benchmark, not the preferred model.

Model-family note

Weighted AutoGluon ensembles top the leaderboard across tracks and feature sets. Tree families supply most of the useful base-model signal; neural nets are considered but do not dominate the final ranks.

Index Construction I: From Score to Portfolio Weight

One score rule, two exclusion regimes

The model produces a probability of CSI next year. A threshold turns that probability into a binary exclusion flag. The flag is then applied to market-cap weights. Two regimes differ only in how long the flag persists after it first fires.

From model score to portfolio weight

Score
probability of CSI next year
from AutoGluon

→

Exclusion flag
1 if score $\geq$ threshold
FPR 1%, 3%, 5%, Youden J

→

Portfolio weight
market-cap $\times$ $(1 - \text{flag})$
renormalized

How the flag evolves over time — a firm crosses the threshold in year 3

Temporary CSI (lockout $L = 3$)	y1	y2	y3 $\star$	y4	y5	y6	y7
Permanent CSI (absorbing)	y1	y2	y3 $\star$	y4	y5	y6	y7

Held — First flag ($\star$) — Excluded

Index Construction II: Four Benchmark Universes

Same exclusion rule, four universes

The revised 11C track applies the score-to-weight pipeline independently inside each CRSP-like universe, producing four screened benchmarks alongside their unfiltered counterparts.

Total Market	**Large Cap**	**Mid Cap**	**Small Cap**
full universe $\geq$ USD 100M	CRSP-like top tier	CRSP-like middle tier	CRSP-like bottom tier

Benchmark construction

- Universe formed from prior-December market capitalization.
- Minimum market cap: USD 100 million.
- Quarterly rebalancing: March, June, September, December.
- Market-cap weights; cost overlays use 0, 5, 10, and 20 bps.

Signal discipline

- Scores are annual for `raw`, `fund`, `latent_raw`, and `raw_plus_latent`.
- Holding-year signal uses information available before the holding year.
- Thresholds calibrated from CV predictions only: Youden, FPR1, FPR3, FPR5.

Index Results: Temporary CSI at 0 bps

OOS benchmark and best strategy by universe; no transaction-cost overlay

Universe	Strategy	Geo ret.	Ann. SD	Sharpe Ratio	Max DD	ES 2.5%	Delta pp
Total	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	0.00pp
Total	AG Base Dataset; Youden/3yr	13.72%	18.28%	0.636	-23.78%	-11.22%	+0.43pp
Large	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	0.00pp
Large	AG Base Dataset; Youden/3yr	14.43%	17.98%	0.678	-23.89%	-10.65%	+0.15pp
Mid	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	0.00pp
Mid	AG Base Dataset; FPR5/5yr	10.16%	20.70%	0.429	-24.46%	-13.85%	+0.51pp
Small	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	0.00pp
Small	AG Exp. Dataset + VAE; Youden/3yr	8.29%	22.65%	0.336	-29.33%	-15.77%	+0.64pp

Temporary CSI conclusion

Without transaction costs, AG Base Dataset wins Total, Large, and Mid Cap. AG Exp. Dataset + VAE wins Small Cap. Delta pp is strategy net geometric return minus the matching benchmark row.

Remark: Source: `best_by_track_index_cost.csv`; Ann. SD and ES 2.5% from model-specific `index_performance_gross_and_net_by_tc.csv`.

Index Results: Temporary CSI at 10 bps

OOS benchmark and best strategy by universe; 10 bps cost rows

Universe	Strategy	Geo ret.	Ann. SD	Sharpe Ratio	Max DD	ES 2.5%	Delta pp
Total	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	0.00pp
Total	AG Base Dataset; Youden/3yr	13.71%	18.28%	0.635	-23.79%	-11.22%	+0.42pp
Large	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	0.00pp
Large	AG Base Dataset; Youden/3yr	14.42%	17.98%	0.677	-23.90%	-10.65%	+0.14pp
Mid	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	0.00pp
Mid	AG Base Dataset; FPR5/5yr	10.05%	20.70%	0.424	-24.46%	-13.85%	+0.39pp
Small	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	0.00pp
Small	AG Exp. Dataset + VAE; Youden/3yr	8.20%	22.64%	0.332	-29.33%	-15.77%	+0.55pp

Temporary CSI transaction-cost read

The 10 bps overlay leaves the temporary CSI winners unchanged. Costs apply to gross buy+sell turnover; benchmark rows stay at the unfiltered market-cap reference with no strategy cost overlay.

Remark: Strategy rows use source-provided 10 bps net performance; expected annual drag is approximately annualized gross turnover $\times$ 0.001. Source: `best_by_track_index_cost.csv`.

Temporary CSI: Error-Cost Decomposition

OOS alpha decomposition for the selected temporary-CSI best strategies

Universe	Best rule	FP cost	FN cost	TP gain	TN gain	Net
Total	AG Base, Youden 3yr	-2.03pp	-0.00pp	+0.08pp	+1.79pp	+0.43pp
Large	AG Base, Youden 3yr	-1.50pp	+0.00pp	-0.00pp	+1.24pp	+0.15pp
Mid	AG Base, FPR5 5yr	+0.05pp	+0.00pp	-0.00pp	+0.27pp	+0.51pp
Small	AG +VAE, Youden 3yr	-4.00pp	-0.02pp	+0.42pp	+3.97pp	+0.64pp

Why temporary CSI outperforms

The strongest gains come from preserving benchmark exposure to correctly retained firms, while false-positive exclusions are the main offset. Small Cap adds visible true-positive gains, but the net result still depends on avoiding too much opportunity cost.

FP and FN are opportunity-cost terms; TP and TN are gains versus the matched benchmark decomposition. Net matches the selected OOS strategy alpha before cost overlays. *Remark:* Source: model-specific `error_cost_decomposition_by_crsp_universe.csv`; matched to `best_by_track_index_cost.csv` by track, universe, model, threshold, lockout, strategy id, and OOS period.

Index Results: Permanent CSI at 0 bps

OOS benchmark and best strategy by universe; permanent-removal rule

Universe	Strategy	Geo ret.	Ann. SD	Sharpe Ratio	Max DD	ES 2.5%	Delta pp
Total	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	0.00pp
Total	AG Exp. Dataset + VAE; FPR5/perm.	13.56%	18.82%	0.615	-24.72%	-11.45%	+0.27pp
Large	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	0.00pp
Large	AG Exp. Dataset + VAE; FPR5/perm.	14.51%	18.38%	0.671	-24.51%	-10.74%	+0.23pp
Mid	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	0.00pp
Mid	AG Base Dataset; FPR5/perm.	10.39%	20.58%	0.441	-24.52%	-13.90%	+0.74pp
Small	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	0.00pp
Small	AG Latent Dataset (VAE); FPR3/perm.	7.98%	23.78%	0.318	-29.26%	-16.35%	+0.32pp

Permanent CSI conclusion

Without transaction costs, AG Exp. Dataset + VAE wins Total and Large Cap; AG Base Dataset wins Mid Cap; AG Latent Dataset (VAE) wins Small Cap.

Remark: Benchmark rows are model-key `bench`, strategy `bench_mw`, period oos. Source: model-specific `index_performance_gross_and_net_by_tc.csv`.

Index Results: Permanent CSI at 10 bps

OOS benchmark and best strategy by universe; 10 bps cost rows

Universe	Strategy	Geo ret.	Ann. SD	Sharpe Ratio	Max DD	ES 2.5%	Delta pp
Total	Benchmark	13.29%	19.03%	0.598	-25.03%	-11.62%	0.00pp
Total	AG Exp. Dataset + VAE; FPR5/perm.	13.55%	18.82%	0.614	-24.72%	-11.45%	+0.26pp
Large	Benchmark	14.28%	18.49%	0.657	-24.79%	-10.84%	0.00pp
Large	AG Exp. Dataset + VAE; FPR5/perm.	14.49%	18.38%	0.670	-24.51%	-10.74%	+0.22pp
Mid	Benchmark	9.65%	20.72%	0.406	-24.58%	-13.83%	0.00pp
Mid	AG Base Dataset; FPR5/perm.	10.28%	20.58%	0.436	-24.52%	-13.90%	+0.62pp
Small	Benchmark	7.65%	23.80%	0.305	-29.05%	-16.12%	0.00pp
Small	AG Latent Dataset (VAE); FPR3/perm.	7.90%	23.78%	0.315	-29.26%	-16.35%	+0.24pp

Permanent CSI transaction-cost read

The 10 bps overlay leaves the permanent CSI winners unchanged. Costs apply to gross buy+sell turnover; benchmark rows stay at the unfiltered market-cap reference with no strategy cost overlay.

Remark: Strategy rows use source-provided 10 bps net performance; expected annual drag is approximately annualized gross turnover $\times$ 0.001. Delta pp compares against the unchanged benchmark row.

Permanent CSI: Error-Cost Decomposition

OOS alpha decomposition for the selected permanent-CSI best strategies

Universe	Best rule	FP cost	FN cost	TP gain	TN gain	Net
Total	AG +VAE, FPR5 perm.	-0.49pp	-0.00pp	+0.01pp	+0.52pp	+0.27pp
Large	AG +VAE, FPR5 perm.	-0.26pp	+0.00pp	+0.00pp	+0.30pp	+0.23pp
Mid	AG Base, FPR5 perm.	-0.55pp	+0.00pp	+0.00pp	+0.90pp	+0.74pp
Small	AG Latent, FPR3 perm.	-0.74pp	-0.00pp	+0.04pp	+1.03pp	+0.32pp

Why permanent CSI outperforms

The permanent rule adds value mostly by keeping exposure to correctly retained firms and avoiding broad over-exclusion. True-positive gains are small because bankrupt/permanent-loss positives are rare, but fixed-FPR rules limit the false-positive cost.

FP and FN are opportunity-cost terms; TP and TN are gains versus the matched benchmark decomposition. Net matches the selected OOS strategy alpha before cost overlays. *Remark:* Source: model-specific `error_cost_decomposition_by_crsp_universe.csv`; matched to `best_by_track_index_cost.csv` by track, universe, model, threshold, permanent rule, strategy id, and OOS period.

Transaction-Cost Robustness

Winner ranking is stable from 0 to 20 bps; costs apply to gross buy+sell turnover

Track	Universe	0 bps winner	20 bps winner	Alpha 0	Alpha 20
Temp.	Total	`fund`	`fund`	+0.43pp	+0.42pp
Temp.	Large	`fund`	`fund`	+0.15pp	+0.15pp
Temp.	Mid	`fund`	`fund`	+0.51pp	+0.51pp
Temp.	Small	`raw_plus_latent`	`raw_plus_latent`	+0.64pp	+0.62pp
Perm.	Total	`raw_plus_latent`	`raw_plus_latent`	+0.27pp	+0.27pp
Perm.	Large	`raw_plus_latent`	`raw_plus_latent`	+0.23pp	+0.23pp
Perm.	Mid	`fund`	`fund`	+0.74pp	+0.74pp
Perm.	Small	`latent_raw`	`latent_raw`	+0.32pp	+0.32pp

Transaction costs do not change the top winner in any of the eight track–index comparisons. The cost drag is economically visible in Mid and Small Cap, but it is not large enough at 20 bps to overturn the ranking. *Remark:*

Alpha is net annualized geometric return minus the corresponding benchmark annualized return. Source: `transaction_cost_robustness_summary.csv`.

Turnover Effect

Best strategies keep their ranking at 10 bps; turnover is gross buy+sell traded weight

Temporary CSI

Universe	Best strategy	Ann. turnover	Geo ret. 0 bps	Geo ret. 10 bps	Delta from TC
Total	AG Base Dataset (Youden 3yr)	15.3%	13.72%	13.71%	-0.01pp
Large	AG Base Dataset (Youden 3yr)	18.8%	14.43%	14.42%	-0.01pp
Mid	AG Base Dataset (FPR5 5yr)	97.7%	10.16%	10.05%	-0.11pp
Small	AG Exp. Dataset + VAE (Youden 3yr)	77.2%	8.29%	8.20%	-0.09pp

Permanent CSI

Universe	Best strategy	Ann. turnover	Geo ret. 0 bps	Geo ret. 10 bps	Delta from TC
Total	AG Exp. Dataset + VAE (FPR5 perm.)	12.2%	13.56%	13.55%	-0.01pp
Large	AG Exp. Dataset + VAE (FPR5 perm.)	17.2%	14.51%	14.49%	-0.01pp
Mid	AG Base Dataset (FPR5 perm.)	97.0%	10.39%	10.28%	-0.12pp
Small	AG Latent Dataset (VAE) (FPR3 perm.)	72.4%	7.98%	7.90%	-0.08pp

Mid and Small Cap exclusions create the largest annualized gross turnover, measured as buy plus sell traded weight, so cost drag is visibly larger there. The 0–20 bps sweep did not change any top winner ranking. *Remark:*

Source: `turnover_summary.csv`, `winner_turnover_summary_20bps.csv`, and `best_by_track_index_cost.csv`.

Threshold Families and Turnover

Threshold-family result at 20 bps

Track	Lead family	Mean alpha
Temp.	Youden	+0.35pp
Perm.	FPR3/FPR5	+0.35/+0.33pp

Temporary CSI benefits from calibrated finite lockouts; 3–5yr exclusions outperform 1yr exclusions. Permanent CSI needs stricter fixed-FPR rules; Youden is negative on average under absorbing exclusion.

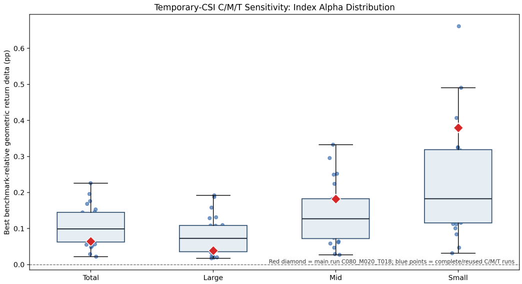
Annualized gross turnover for 20 bps winners

Universe	Temp.	Perm.
Total	15.3%	12.2%
Large	18.8%	17.2%
Mid	97.7%	97.0%
Small	77.2%	72.4%

Turnover matters most in Mid and Small Cap. At 20 bps, the annual return drag is still modest: roughly 1.1pp in Mid Cap and below 0.9pp in Small Cap winners.

Remark: Sources: `threshold_family_summary_20bps.csv` and `winner_turnover_summary_20bps.csv`.

Sensitivity: Main Run Versus C/M/T Grid



Key results: best alpha distribution

Univ.	Runs	Main	Median	Range
Total	24	0.06	0.10	0.02–0.23
Large	24	0.04	0.07	0.02–0.19
Mid	24	0.18	0.13	0.03–0.33
Small	24	0.38	0.18	0.03–0.66

Values are benchmark-relative geometric return deltas in percentage points.

- The accepted main run sits inside the completed-run distribution.
- Main-run alpha is below the median for Total/Large and above the median for Mid/Small.
- Sensitivity changes magnitude more than direction: completed runs remain positive across all four universes.
- Three blocked partial C/M/T configs remain excluded from completed-run conclusions.

Remark: Sources: `chart1_sensitivity_stability_distribution.png`; `universe_stability_summary.csv`; `sensitivity_index_stability_table.csv`; AE-SENS-CHART-001R interpretation.

Remaining Robustness Work

What is complete vs. still future work

Temporary-CSI C/M/T sensitivity is complete locally and supports the presentation robustness narrative. The remaining items below are extensions or validation work, not completed results.

Permanent-CSI sensitivity grid

Repeat the C/M/T-style robustness logic for the permanent response track before claiming permanent-label sensitivity.

Benchmark extensions

Minimum-volatility and quality/risk-scaling benchmarks remain future comparator work.

Probability-weighted strategy

Active CSI-probability weighting is not yet implemented; current results use thresholded exclusions.

Residual terminal-failure diagnostic

Recovered bankrupt/insolvent firm diagnostics remain a follow-up, not a completed robustness result.

Remark: The final index suite and the temporary sensitivity grid answer different questions and are kept separate in the presentation.

Appendix A1: Delisting and Bankruptcy Detection

Indicator group	CRSP firms	Detected	Rate	Positive rows followed
Bankruptcy range 572-574	629	545	86.65%	2,757
Bankruptcy / declared insolvent 574	548	513	93.61%	2,502
Adverse 400-490, 550-585	5,654	3,676	65.02%	7,000

Why the override is additive

The CRSP signal is not an independent default label. It only turns a CSI-like trigger into a positive `dynamic_csi` event when the firm fails before the 18-month confirmation clock can fully resolve.

Appendix A2: Revised Annual Prevalence

Annual label metric	Count
Temporary positive label cells	8,517
Temporary unresolved cells ($y = NA$)	8,674
Permanent positive label cells	6,258
Permanent unresolved cells ($y = NA$)	834

Current CRSP 572–574 overlap

Metric	Count
CRSP 572–574 firms in sample	629
Detected by revised temporary CSI	545
Missed by revised temporary CSI	84
Detection rate	86.65%

The remaining misses are not treated as current pipeline errors. Under the additive rule, CRSP bankruptcy codes only validate an already triggered CSI event within the timing window; they are not standalone default labels.

Methodological status

The current overlap audit supports the 545/84/86.65% story. A full reason breakdown for all 84 CRSP 572–574 misses would require a separate diagnostic; the current slide treats them as exclusions under the retained trigger-and-window definition.

Appendix A3: Cleaned Label Counts

CSI type	Split	Obs. rows	$y=0$	$y=1$	$y=NA$	Labelled	Prev.
Temporary	Full	188,460	171,269	**8,517**	8,674	179,786	4.74%
Temporary	Train	143,173	136,630	6,543	0	143,173	4.57%
Temporary	Test	18,111	17,307	804	0	18,111	4.44%
Temporary	OOS	27,176	17,332	1,170	8,674	18,502	6.32%
Permanent	Full	188,460	181,368	6,258	834	187,626	3.34%
Permanent	Train	143,173	137,794	5,379	0	143,173	3.76%
Permanent	Test	18,111	17,511	542	58	18,053	3.00%
Permanent	OOS	27,176	26,063	337	776	26,400	1.28%

Read

Both targets remain rare after the observable-scaffold cleanup. $y = NA$ rows stay in canonical artifacts but are excluded from supervised training and label-based evaluation.

Source: Labelled rows $= y = 0 + y = 1$; prevalence $= y = 1/$labelled rows.

Appendix A4: Descriptive Statistics – Market Cap

Market cap, USD millions

CSI type	Split	y	Mean	Median	P25	P75
Temporary	CV	0	4,117.4	408.1	87.3	1,924.5
Temporary	CV	1	613.4	85.1	30.0	255.5
Temporary	Full	0	4,864.6	350.9	67.2	1,903.7
Temporary	Full	1	574.6	68.7	23.3	228.0
Permanent	CV	0	3,452.1	354.5	85.2	1,551.6
Permanent	CV	1	447.3	79.6	30.3	221.6
Permanent	Full	0	3,558.3	249.3	57.7	1,232.6
Permanent	Full	1	496.5	66.2	22.7	219.0

Read

CSI-positive firms are much smaller across both definitions. In the full sample, median market cap is 68.7 for temporary positives versus 350.9 for non-positives, and 66.2 for permanent positives versus 249.3 for non-positives.

Source: Market cap is USD millions.

Appendix A5: Descriptive Statistics – Other Medians

CSI type	Split	y	Ann. ret.	Assets	Sales	ROA	M/B
Temporary	CV	0	6.1%	640.9	326.7	1.7%	1.82
Temporary	CV	1	-24.9%	86.1	50.2	-14.2%	1.51
Temporary	Full	0	4.9%	565.3	295.2	1.6%	1.81
Temporary	Full	1	-35.0%	77.5	46.8	-18.8%	1.59
Permanent	CV	0	6.0%	567.8	270.2	1.7%	1.82
Permanent	CV	1	-28.3%	78.8	44.4	-15.8%	1.46
Permanent	Full	0	4.2%	395.3	191.9	1.5%	1.80
Permanent	Full	1	-37.5%	64.6	35.1	-19.3%	1.52

Read

The positive class has sharply lower realized returns, smaller operating scale, and negative profitability. The contrast strengthens in the full sample for both temporary and permanent CSI.

Source: Assets and sales are USD millions.

Appendix A6: CV Folds, Leakage Controls, and Training Metric

Expanding-window CV

Fold	Training years	Validation years
1	Initial training block	No validation; used to seed expanding window
2	Up to 1998	1999-2004
3	Up to 2004	2005-2009
4	Up to 2009	2010-2015

Training / selection metric

- **AutoGluon:** yes, model ranking uses AP via `EVAL_METRIC = "average_precision"` passed to `TabularPredictor(..., eval_metric=EVAL_METRIC)` in `09C_AutoGluon.py`.
- **XGBoost:** binary-logistic training uses `eval_metric="aucpr"` in `09_Train.R`; reported AP is computed from predicted probabilities.
- Split before preprocessing.
- Train-fitted winsorization, imputation, and PIT only.
- CV AP/AUC/R@FPR metrics are averaged per fold; OOS is reserved for index construction, not model selection.

Appendix A7: Feature Family Inventory

Family	Examples / purpose
Fundamentals	Balance sheet, income statement, profitability, leverage, liquidity, accounting dynamics.
Market and price	Returns, drawdowns, volatility, momentum, market capitalization, price-path information.
Macro	FRED macro series and macro interaction terms.
Latent	VAE latent factors from fundamentals or raw features plus reconstruction error.
Metadata excluded	Labels, event dates, confirmation dates, censoring flags, and diagnostics are retained for audit but excluded from model features.

Appendix A8: VAE Architecture Details

Component	Setting
Encoder	$n\_features \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow z\_mean, z\_log\_var$
Latent dimension	24
Decoder	$z \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow n\_features$
Loss	Reconstruction + β KL + γ supervised CSI classification loss
Parameters	$\beta = 1.0$, $\gamma = 0.1$, early stopping patience 15
Status	Weakly supervised on training rows; test and OOS labels are not used for VAE fitting.

Appendix A9: Temporary CSI CV/Test/OOS Model Metrics

Full model-suite threshold metrics Train/CV, test, and OOS; AP/AUC/FPR recall.

Split	Model	AP	AUC	R@FPR1	R@FPR3	R@FPR5	Brier
CV	AG Expanded Dataset	0.2046	0.8635	0.0917	0.2344	0.3415	0.0371
CV	AG Base Dataset	0.1923	0.8446	0.0830	0.2206	0.3229	0.0374
CV	AG Latent Dataset (VAE)	0.1659	0.8454	0.0670	0.1706	0.2735	0.0380
CV	AG Exp. Dataset + VAE	**0.2114**	**0.8667**	**0.0981**	**0.2478**	**0.3581**	**0.0368**
Test	AG Expanded Dataset	**0.1985**	**0.8766**	**0.0821**	**0.2002**	**0.2998**	**0.0389**
Test	AG Base Dataset	0.1656	0.8538	0.0585	0.1517	0.2413	0.0392
Test	AG Latent Dataset (VAE)	0.1501	0.8438	0.0410	0.1294	0.2264	0.0394
Test	AG Exp. Dataset + VAE	0.1864	0.8741	0.0634	0.1779	0.2749	0.0394
OOS	AG Expanded Dataset	0.3084	**0.8961**	0.0915	0.2547	0.4000	0.0498
OOS	AG Base Dataset	0.2526	0.8686	0.0598	0.1795	0.3017	0.0526
OOS	AG Latent Dataset (VAE)	0.2813	0.8626	0.1009	0.2274	0.3556	0.0518
OOS	AG Exp. Dataset + VAE	**0.3152**	0.8950	**0.1051**	**0.2632**	**0.4128**	**0.0494**

Remark: Source: `03_Data_Output/6_ModelSuite/comparison/AE-MODEL-SUITE-007_model_suite_metrics_long.csv`; raw comparator from `raw/derived_metrics/raw_complete_threshold_metrics.csv`.

Appendix A10: Permanent CSI CV/Test/OOS Model Metrics

Full model-suite threshold metrics Train/CV, test, and OOS; AP/AUC/FPR recall.

Split	Model	AP	AUC	R@FPR1	R@FPR3	R@FPR5	Brier
CV	AG Expanded Dataset	0.1854	**0.8757**	0.1098	**0.2607**	**0.3697**	0.0297
CV	AG Base Dataset	0.1785	0.8677	0.1033	0.2379	0.3469	0.0299
CV	AG Latent Dataset (VAE)	0.1426	0.8460	0.0732	0.1960	0.2908	0.0308
CV	AG Exp. Dataset + VAE	**0.1883**	0.8719	**0.1196**	0.2578	0.3680	**0.0297**
Test	AG Expanded Dataset	**0.1416**	0.8810	**0.0664**	0.1808	0.3007	0.0284
Test	AG Base Dataset	0.1289	0.8627	0.0627	0.1697	0.2620	0.0293
Test	AG Latent Dataset (VAE)	0.1115	0.8496	0.0424	0.1513	0.2546	**0.0280**
Test	AG Exp. Dataset + VAE	0.1415	**0.8838**	0.0609	**0.2011**	**0.3155**	0.0283
OOS	AG Expanded Dataset	0.0323	0.8081	0.0000	0.0119	0.0623	0.0211
OOS	AG Base Dataset	0.0281	0.7798	0.0000	0.0030	0.0297	0.0232
OOS	AG Latent Dataset (VAE)	**0.0482**	**0.8337**	**0.0504**	**0.1484**	**0.2226**	**0.0166**
OOS	AG Exp. Dataset + VAE	0.0311	0.8032	0.0000	0.0208	0.0653	0.0207

Remark: Source: `03_Data_Output/6_ModelSuite/comparison/AE-MODEL-SUITE-007_model_suite_metrics_long.csv`; raw comparator from `raw/derived_metrics/raw_complete_threshold_metrics.csv`.

AutoGluon selection evidence

Every final feature-set/track run selected a `WeightedEnsemble_L2` as the top leaderboard model. The useful base signal is mostly tree-based; neural nets were considered but did not dominate.

Track	Feature set	Best family	Val. score	Fit sec.
Temporary	AG Expanded Dataset	WeightedEnsemble	0.2307	280
Temporary	AG Base Dataset	WeightedEnsemble	0.2203	83
Temporary	AG Latent Dataset (VAE)	WeightedEnsemble	0.1953	146
Temporary	AG Exp. Dataset + VAE	WeightedEnsemble	**0.2314**	203
Permanent	AG Expanded Dataset	WeightedEnsemble	0.2010	126
Permanent	AG Base Dataset	WeightedEnsemble	0.1929	229
Permanent	AG Latent Dataset (VAE)	WeightedEnsemble	0.1505	191
Permanent	AG Exp. Dataset + VAE	WeightedEnsemble	**0.2041**	291

Remark: Source: `AE-MODEL-SUITE-007_model_family_winners.csv`.

Appendix A12: Model Metric Caveats and OOS Robustness

Comparability

The raw comparator is the revised-dataset optional-library AutoGluon rerun from AE-VALIDATE. Complete local threshold metrics were later materialized under `03_Data_Output/6_ModelSuite/raw/derived_metrics`, so the presentation uses AP, AUC, Brier, and R@FPR1/3/5 consistently across feature sets.

- AP and fixed-FPR recall are emphasized because CSI positives are rare.
- AUC remains useful as a ranking diagnostic but can mask low-prevalence false-positive economics.
- AG Exp. Dataset + VAE is the main enhanced feature-set challenger for temporary CSI and permanent test/CV.
- AG Latent Dataset (VAE) is retained for permanent OOS robustness, not as a universal replacement.
- Economic superiority is not claimed from model metrics alone; index-construction slides test benchmark-relative performance.

Appendix A13: Final Index Grid Contract

Completed grid

- Models: `raw` (AG Expanded Dataset), `fund` (AG Base Dataset), `latent_raw` (AG Latent Dataset), `raw_plus_latent` (AG Exp. Dataset + VAE).
- Tracks: temporary CSI and permanent CSI.
- Universes: Total Market, Large Cap, Mid Cap, Small Cap.
- Thresholds: Youden, FPR1, FPR3, FPR5.

Portfolio rules

- Temporary CSI: 1-, 2-, 3-, and 5-year exclusion lockouts.
- Permanent CSI: absorbing permanent removal after first active signal.
- Costs: 0, 5, 10, and 20 bps on gross turnover.
- Turnover output is populated for every model–track–universe cell.

Presentation interpretation

The index section uses OOS economic performance. Model metrics identify useful prediction signals; index construction tests whether those signals improve benchmark-relative returns after thresholding, exclusions, turnover, and transaction costs.

Remark: Sources: `comparison/full_grid_manifest.csv`; `comparison/best_by_track_index_cost.csv`; `comparison/turnover_summary.csv`.

Appendix A14: 20 bps Winners With Benchmarks

Best strategy by track and universe at 20 bps

Track	Universe	Winner	Rule	Net ret.	Bench.	Alpha
Temp.	Total	`fund`	Youden / 3yr	13.70%	13.28%	+0.42pp
Temp.	Large	`fund`	Youden / 3yr	14.40%	14.25%	+0.15pp
Temp.	Mid	`fund`	FPR5 / 5yr	9.93%	9.42%	+0.51pp
Temp.	Small	`raw_plus_latent`	Youden / 3yr	8.11%	7.49%	+0.62pp
Perm.	Total	`raw_plus_latent`	FPR5 / perm.	13.54%	13.28%	+0.27pp
Perm.	Large	`raw_plus_latent`	FPR5 / perm.	14.48%	14.25%	+0.23pp
Perm.	Mid	`fund`	FPR5 / perm.	10.16%	9.42%	+0.74pp
Perm.	Small	`latent_raw`	FPR3 / perm.	7.82%	7.49%	+0.32pp

VAE/non-raw variants beat raw in 5 of 8 track–index cases at 20 bps. The market-weighted benchmark is shown as the return reference and does not receive a strategy cost overlay. *Remark:* Source:

`presentation_headline_tables.md`; final claim from `AE-INDEX-SUITE-009_Closeout_Report.md`.

Appendix A15: Error Decomposition and Index Source Paths

Base folders: `03_Data_Output/7_IndexConstructionValidation/nonraw_index_suite/`

- Headline winners: `final_tables/headline_winners_20bps.csv`
- Presentation tables: `final_tables/presentation_headline_tables.md`
- Transaction costs: `comparison/transaction_cost_impact.csv`; `final_tables/transaction_cost_robustness_summary.csv`
- Best-by-cost grid: `comparison/best_by_track_index_cost.csv`
- Threshold-family grid: `comparison/best_by_track_index_threshold_family.csv`
- Strategy performance fields: model-specific `index_performance_gross_and_net_by_tc.csv`
- Threshold families: `final_tables/threshold_family_summary_20bps.csv`
- Turnover: `comparison/turnover_summary.csv`; `final_tables/winner_turnover_summary_20bps.csv`
- Error-cost decomposition: selected `fund`, `raw_plus_latent`, and `latent_raw` `error_cost_decomposition_by_crsp_universe.csv` files mapped exactly in `SLIDE_DATA_SOURCES.md`.

Source discipline

All claims in the index-result slides are sourced from the completed AE-INDEX-SUITE grid. Sensitivity-analysis results are intentionally not finalized in this section.

Appendix A16: Transaction-Cost Robustness

Top winner did not change between 0 and 20 bps

Track	Universe	Winner 0 bps	Winner 20 bps	Alpha 0	Alpha 20
Temp.	Total	`fund`	`fund`	+0.43pp	+0.42pp
Temp.	Large	`fund`	`fund`	+0.15pp	+0.15pp
Temp.	Mid	`fund`	`fund`	+0.51pp	+0.51pp
Temp.	Small	`raw_plus_latent`	`raw_plus_latent`	+0.64pp	+0.62pp
Perm.	Total	`raw_plus_latent`	`raw_plus_latent`	+0.27pp	+0.27pp
Perm.	Large	`raw_plus_latent`	`raw_plus_latent`	+0.23pp	+0.23pp
Perm.	Mid	`fund`	`fund`	+0.74pp	+0.74pp
Perm.	Small	`latent_raw`	`latent_raw`	+0.32pp	+0.32pp

Costs are applied at 0, 5, 10, and 20 bps. The 20 bps case is conservative enough for presentation, but not high enough to reverse the headline rankings. *Remark:* Source: `transaction_cost_robustness_summary.csv`.

Appendix A17: Threshold Family and Turnover Summary

Mean best alpha by threshold family

Track	Threshold	Mean alpha
Temp.	Youden	+0.35pp
Temp.	FPR3	+0.19pp
Temp.	FPR5	+0.17pp
Temp.	FPR1	+0.12pp
Perm.	FPR3	+0.35pp
Perm.	FPR5	+0.33pp
Perm.	FPR1	+0.14pp
Perm.	Youden	-0.20pp

Mean annualized turnover for winners

Universe	Temp.	Perm.
Total	15.3%	12.2%
Large	18.8%	17.2%
Mid	97.7%	97.0%
Small	77.2%	72.4%

Temporary CSI is strongest with Youden and longer finite lockouts; permanent CSI is strongest with FPR3/FPR5. Turnover is concentrated in Mid and Small Cap, where exclusions shift more of the investable universe. *Remark:* Sources: `threshold_family_summary_20bps.csv`; `winner_turnover_summary_20bps.csv`.

Appendix A18: Temporary CSI Sensitivity Detail

Selected configurations from the 27-run temporary-CSI sensitivity grid

Role	Run ID	Pos.	Prev.	OOS AP	OOS AUC	11C TM alpha	Status
Composite	C090_M000_T012	12,761	7.02%	0.557	0.920	+0.169pp	complete
AP winner	C060_M000_T012	31,596	17.49%	0.566	0.864	+0.052pp	reused
11C winner	C090_M020_T018	6,035	3.34%	0.274	0.911	+0.226pp	complete
Baseline	C080_M020_T018	8,517	4.74%	0.308	0.896	+0.065pp	reused

Blocked partial cases

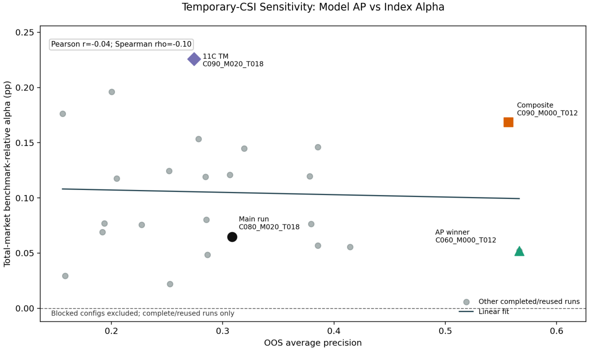
The grid represents 27 C/M/T configurations: 24 are complete or reused, and three are `blocked_partial`: `C080_M000_T012`, `C080_M000_T018`, and `C060_M020_T028`. Blocked cases are represented in manifests but excluded from completed-run conclusions.

Source interpretation

C/M/T sensitivity tests label-threshold robustness for temporary CSI. It supports the robustness narrative but does not alter the final accepted label definition used by AE-INDEX-SUITE. In the accepted-label temporary index grid, 0/5/10/20 bps transaction-cost overlays do not change the winner set: Base Dataset wins Total/Large/Mid and Exp. Dataset + VAE wins Small. Costs are applied to annualized gross buy+sell turnover.

Remark: Sources: local `sensitivity_cmt_model_summary.csv`; `sensitivity_cmt_index_summary.csv`; `temporary_index_transaction_cost_lines.csv`; `temporary_blocked_config_disclosure.csv`.

Appendix A19: Model AP Versus Index Alpha



High AP and high index alpha are related but not identical objectives.

Model versus portfolio objective

- AP and index alpha are related robustness diagnostics, but they are not the same objective.
- The AP winner and total-market alpha winner differ across C/M/T settings.
- The continuity baseline remains defensible, but it is not top-ranked.
- This supports presenting sensitivity as a robustness check rather than a replacement for the accepted main run.

Remark: Sources: `chart2_model_vs_index_sensitivity_scatter_refined.png`; local `sensitivity_cmt_model_summary.csv`; `sensitivity_cmt_index_summary.csv`; AE-SENS-CHART-001R interpretation.

Appendix A20: Future Robustness and Extensions

Not completed results

The items below are explicitly future validation or extensions. They should not be read as completed empirical findings in this deck.

Item	Purpose	Current status
Permanent-CSI sensitivity grid	Test label-threshold robustness for the permanent response track.	Planned / not completed
Minimum-volatility benchmark	Check whether CSI screens add value beyond a low-risk index comparator.	Planned / not completed
Quality/risk-scaling benchmark	Compare exclusions with quality or risk-scaled benchmark construction.	Planned / not completed
Active CSI-probability weighting	Use CSI probabilities continuously rather than thresholded exclusions.	Planned / not completed
Recovered bankrupt/insolvent diagnostic	Decompose remaining terminal-failure misses and recovery cases.	Planned / not completed

Remark: Source status is mapped in `SLIDE_DATA_SOURCES.md`; these are future-work notes, not result claims.

Appendix A21: Slide-to-Source Audit Map

Audit artifact

The complete slide-to-source map is stored beside this June source deck:
`06_Presentations/02_FinalPresentation/Necessary/FinalPresentation_June/SLIDE_DATA_SOURCES.md`.

- Dataset and label slides map to AE-PANEL evidence and descriptive-statistics CSVs.
- Model slides map to AE-MODEL-SUITE comparison files and threshold metrics.
- Index slides map to AE-INDEX-SUITE final-table and comparison outputs.
- Sensitivity slides map to local AE-SENS manifests and comparison artifacts.
- Conceptual or future-work slides are marked as such rather than linked to empirical result files.

Remark: This appendix frame is a navigation pointer; the source map contains per-frame paths and source type.

Bibliography I

Edward I. Altman. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4):589–609, 1968.

John Y. Campbell, Jens Hilscher, and Jan Szilagyi. In search of distress risk. *The Journal of Finance*, 63(6):2899–2939, 2008.

Michael Cembalest. The agony & the ecstasy: The risks and rewards of a concentrated stock position. Eye on the market special edition, J.P. Morgan Asset Management, 2014. September 2, 2014.

Stephen Penman and Francesco Reggiani. Fundamentals of value versus growth investing and an explanation for the value trap. *Financial Analysts Journal*, 74 (4):103–119, 2018. doi: 10.2469/faj.v74.n4.6.

Tyler Shumway. The delisting bias in CRSP data. *The Journal of Finance*, 52(1): 327–340, 1997.

Tyler Shumway. Forecasting bankruptcy more accurately: A simple hazard model. *The Journal of Business*, 74(1):101–124, 2001.

Zaki Tewari, Michal Galas, and Philip Treleaven. Predicting catastrophic stock implosions: A machine learning approach. Technical report, University College London, 2024. Working Paper.